MTH 202 Chapter 2: Vector-Valued Functions

2.3 Arc Length & Change of Parameter

<u>Recall:</u> In calculus II you probably were introduced to the formula $\displaystyle\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt$ for the length of a parametric curve in two dimensions. Since the distance formula in three dimensions is very closely related to the distance formula in two dimensions, this formula extends nicely in the 3-dimensional case. However, before we present this formula we introduce some terminology.

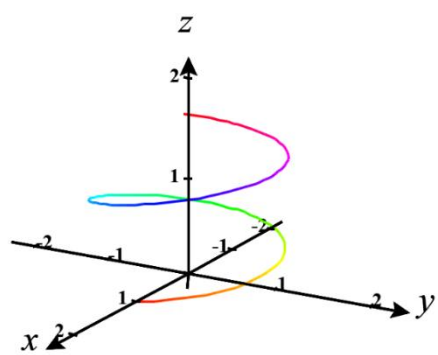
Definition: A parameterization $\mathbf{r}(t)$ is called <u>smooth</u> on an interval *I* if $\mathbf{r}'$ is continuous and $\mathbf{r}'(t)\neq\mathbf{0}$ for any *t* on *I*. A curve is <u>smooth</u> if it has a smooth parameterization. A smooth curve has no sharp corners or cusps.

Theorem: Suppose that $\mathbf{r}(t)=\langle x(t),y(t),z(t)\rangle$ is a smooth vectored-value function for $a\le t\le b$, where the curve is only traversed one time as *t* goes from *a* to *b* then

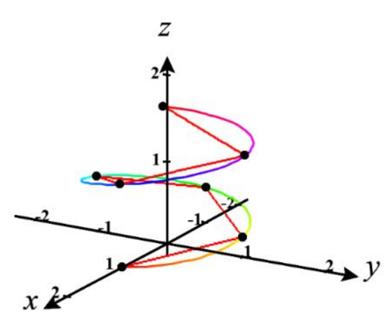
$$L=\int_a^b\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2+\left(\frac{dz}{dt}\right)^2}\,dt=\int_a^b\left\|\mathbf{r}'(t)\right\|dt\,.$$

Partial Proof:

Suppose we have a smooth vector-valued function given by $\mathbf{r}(t)$ where $a\le t\le b$.
We will begin by dividing the interval $[a,b]$ into *n* subintervals $[t_{i-1},t_i]$ of equal length Δt and connect the endpoints of these subintervals with line segments to approximate the length of the curve.



We measure the length of any given line segment by creating a vector between the endpoints, which we will call $\mathbf{r}_i$, and calculating its magnitude. From this we obtain that $L_i=\|\mathbf{r}_i\|=\sqrt{\Delta x_i^{\,2}+\Delta y_i^{\,2}+\Delta z_i^{\,2}}$.



Therefore, we estimate the total length of the vector function to be

$$L\approx\sum_{i=1}^{n}L_i=\sum_{i=1}^{n}\|\mathbf{r}_i\|=\sum_{i=1}^{n}\sqrt{\Delta x_i^{\,2}+\Delta y_i^{\,2}+\Delta z_i^{\,2}}\,.$$

With a much more complicated argument it can be shown that the FTOC and the MVT can be used to establish that

$$\sqrt{\Delta x_i^{\,2}+\Delta y_i^{\,2}+\Delta z_i^{\,2}}=\sqrt{x'(t_i)^2+y'(t_i)^2+z'(t_i)^2}\,\Delta t \quad \text{(this argument is beyond the scope of this course)}$$

Therefore $L \approx \sum_{i=1}^{n}\|\mathbf{r}_i\| = \sum_{i=1}^{n}\sqrt{x'(t_i)^2+y'(t_i)^2+z'(t_i)^2}\,\Delta t$.

We therefore expect that as $n\to\infty$ we will obtain the exact arc length.

So, we have $L=\lim_{n\to\infty}\sum_{i=1}^{n}\|\mathbf{r}_i\|=\lim_{n\to\infty}\sum_{i=1}^{n}\sqrt{x'(t_i)^2+y'(t_i)^2+z'(t_i)^2}\,\Delta t=\int_a^b\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2+\left(\frac{dz}{dt}\right)^2}\,dt=\int_a^b\|\mathbf{r}'(t)\|\,dt$

<u>**Example 1:**</u> Find the length of the curve determined by $\mathbf{r}(t)=\cos(2t)\mathbf{i}+\sin(2t)\mathbf{j}+t\mathbf{k}$ from the point $(1,0,0)$ and $\left(-1,0,\frac{\pi}{2}\right)$.

We first note that the curve is traversed once as t increases from 0 to $\frac{\pi}{2}$.

We note that $\mathbf{r}'(t)=-2\sin(2t)\mathbf{i}+2\cos(2t)\mathbf{j}+\mathbf{k}$.

So, $\mathbf{r}'$ is clearly continuous from 0 to $\frac{\pi}{2}$ and also $\mathbf{r}'(t)\neq\mathbf{0}$ on 0 to $\frac{\pi}{2}$.

So, the desired length is:

$$L=\int_0^{\pi/2}\|\mathbf{r}'(t)\|dt=\int_0^{\pi/2}\sqrt{(-2\sin(2t))^2+(2\cos(2t))^2+1^2}\,dt=\int_0^{\pi/2}\sqrt{5}dt=\frac{\sqrt{5}\pi}{2}\ .$$

Definition: Suppose a curve C is given by $\mathbf{r}(t)=\langle f(t),g(t),h(t)\rangle$ where $a\le t\le b$ where $\mathbf{r}'$ is continuous and C is traversed exactly once as t increases from a to b. We define the <u>arc length function of C</u> to be

$$s(t)=\int_a^t\|\mathbf{r}'(u)\|\,du \ \text{ for } t \text{ in } [a,b].$$

Note: Intuitively the arc length function of C takes in t and spits out the length of the part of C traversed from a to t.

Remark: By the fundamental theorem of calculus $\frac{ds}{dt}=\|\mathbf{r}'(t)\| \to ds=\|\mathbf{r}'(t)\|dt$. Therefore, as we will see later in Chapter 5, the arc length integral is sometimes written as $L=\int_a^b ds$.

<u>Example 2:</u> Suppose C is given by $\mathbf{r}(t)=\cos(2t)\mathbf{i}+\sin(2t)\mathbf{j}+t\mathbf{k}$ where $0\le t\le\frac{\pi}{2}$. Find the arc length function of C.

Since this is the curve from Example 1 we already checked the necessary conditions. So, by definition:

$$s(t)=\int_0^t\|\mathbf{r}'(u)\|du=\int_0^t\sqrt{5}du=\left(\sqrt{5}\right)t \text{ where } 0\le t\le\frac{\pi}{2}.$$

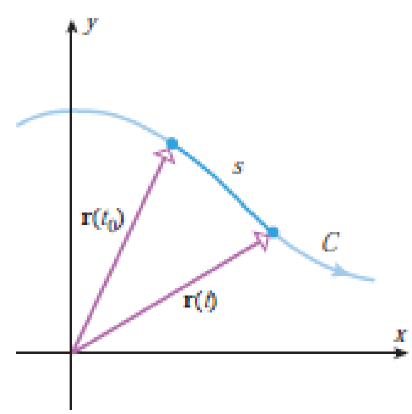
In the past several sections we have communicated curves in 3-space using parametric equations which used a parameter of t that we commonly thought of as representing time. However, it can be helpful to change parameters so that we can measure motion along a curve not just by some value of time. Specifically, it will be helpful to measure motion along a curve in terms of the distance we have travelled from some fixed starting point, that is to say, using arc length.

Definition: Suppose a curve C is given by $\mathbf{r}(t)=\langle f(t),g(t),h(t)\rangle$ with $a\le t\le b$ where $\mathbf{r}$ is a smooth vector-valued function and C is traversed exactly once as t increases from a to b. If from the arc length function for C we can solve for t to obtain t as a function of s, then the <u>arc length parameterization of C</u> is $\mathbf{r}(t(s))$.

Steps to Obtaining an Arc Length Parameterization for a Curve:

1. Find the arc length function of C, $s(t)$.
2. Use $s(t)$ to solve for t in terms of s (this may not be possible). Then, you have t as a function of s.
3. Simply use composition of functions to obtain your arc length parameterization: $\mathbf{r}(t(s))$.

Remark: The power in the arc length parameterization is that if your plug a number like 2 into it, it will spit out a position vector whose terminal point is the point at which a particle which started its movement at the initial point of C is located after it has traveled 2 units along C.



Example 3: Suppose C is given by $\mathbf{r}(t) = \cos(2t)\mathbf{i} + \sin(2t)\,\mathbf{j} + t\mathbf{k}$ where $0 \le t \le \frac{\pi}{2}$.

a) Find an arc length parameterization of C.

We know from the previous example that $s(t) = \left(\sqrt{5}\right)t$. This implies that $t = \frac{s}{\sqrt{5}}$, and the arc length parameterization is $\mathbf{r}(t(s)) = \cos\left(\frac{2s}{\sqrt{5}}\right)\mathbf{i} + \sin\left(\frac{2s}{\sqrt{5}}\right)\mathbf{j} + \left(\frac{s}{\sqrt{5}}\right)\mathbf{k}$.

b) Plug $s = \frac{\sqrt{5}\pi}{2}$ into the parameterization you found in (a). Look back in your notes to see why the result that you get is awesome.

We calculate $\mathbf{r}\left(t\left(\frac{\sqrt{5}\pi}{2}\right)\right) = \cos(\pi)\mathbf{i} + \sin(\pi)\mathbf{j} + \left(\frac{\pi}{2}\right)\mathbf{k} = -\mathbf{i} + \left(\frac{\pi}{2}\right)\mathbf{k}$, and we know from the second to last example that the terminal point of this position vector is where one must travel on the curve from $(1,0,0)$ to get a length of $\frac{\sqrt{5}\pi}{2}$!